

HANIEH HAERI

San Ramon, CA • +1 925-768-0299 • hhaeri0911@gmail.com
hhaeri.github.io/portfolio • linkedin.com/in/hanieh-haeri-9319b024

TECHNICAL SKILLS

- **Generative AI & Deep Learning:** Large Language Models (LLMs), RAG, LangChain, ReAct Frameworks, OpenAI/Google Gemini APIs, PyTorch, Keras, TensorFlow, LSTM.
- **Classical Machine Learning & NLP:** Classification, Regression, Clustering, Anomaly Detection, Natural Language Processing (NLP), Sentiment Analysis, Text Mining, Scikit-Learn.
- **Data Engineering, Cloud & MLOps:** Google Cloud Platform (GCP), BigQuery, Spark, Databricks, Pipeline Automation, Model Deployment, Automated ETL, DuckDB, SQL.
- **Programming & Libraries:** Python, R, Scala, Java, MATLAB, NumPy, Pandas, Shapely.
- **Geospatial & Visualization:** GeoPandas, Folium, Carto, GIS Analysis, Tableau, Matplotlib.
- **Methodologies:** Customer Analytics, Predictive Modeling, Survey Analysis, Churn Prediction, Customer Segmentation, Spatial Analytics, Causal Inference, Time-Series Forecasting, A/B Testing, Hypothesis Testing.

PROFESSIONAL EXPERIENCE

Founder & CTO | *Stealth AI Startup (HydroAgent)* 2025 - Present

- Developed a high-fidelity AI-powered spatial analytics platform designed to modernize groundwater resource management and forecasting using LangChain and Google Gemini.
- Mitigated LLM hallucinations by engineering a custom Retrieval-Augmented Generation (RAG) pipeline utilizing autonomous agentic orchestration and deterministic tool-calling. This processed both structured tabular data (via CNRA APIs and datastores) and large volumes of unstructured text data (annual reports and Groundwater Sustainability Plans), grounding AI responses in real-world geospatial data to ensure high model transparency and reliability for end-users.
- Automated production data pipelines and designed a high-speed analytical backend using DuckDB and GeoPandas for real-time compliance checking across thousands of monitoring sites statewide.
- Architected interactive spatial visualizations using Carto and Folium to provide streamlined, visual tools for end-users, delivering data-validated regional insights for cross-functional stakeholders.

Adjunct Faculty of Business Analytics & Data Science | *Saint Mary's College of California* 2025 - Present

- Design and instruct graduate-level courses in Business Analytics, Applied Python Programming, Machine Learning, and Natural Language Processing (NLP).
- Synthesize complex technical information, algorithms, and data insights into clear, digestible presentations for diverse, non-technical and technical audiences.

Machine Learning Researcher | *University of California – San Francisco, CA*

2023 - 2024

- Enhanced a Generative model using a Variational Autoencoder (VAE) architecture. Extracted interpretable features from large unstructured image datasets to drive AI-assisted diagnoses in medical imaging.
- Prioritized improved model transparency and reliability so medical professionals could trust the outputs, translating complex data into actionable clinical insights.

Engineer | *Montgomery & Associates, Oakland, CA*

2014 - 2022

- Automated end-to-end data pipelines for diverse, large-scale hydrogeologic datasets, integrating Python and SQL workflows. Reduced processing bottlenecks by 60% and accelerated team decision-making.
- Developed and deployed production-grade Machine Learning models to forecast resource trends, providing critical technical support and optimizing operational outcomes.
- Translated complex technical insights into actionable recommendations and streamlined interactive dashboards, directly influencing client strategic planning and improving the customer experience.

SELECTED PROJECTS

Causal Machine Learning for Coupon Campaign Optimization

- Leveraged Double ML for causal inference in marketing data to optimize customer experiences and targeted marketing. Identified high-impact customer segments and translated complex data into actionable insights, increasing campaign ROI by 48%.

Semantic Segmentation for Extracting Geologic Features from Historic Topographic Maps

- Developed a UNet-based Semantic Segmentation model in TensorFlow to analyze unstructured image data. This automated feature extraction from historic maps reduced manual processing efforts by 54%, accelerating project timelines and improving the quality of the visual data products delivered to clients.

Timeseries Forecasting of Daily COVID Cases in School Age Population

- Built ARIMA and LSTM time series models to forecast daily COVID-19 cases among California school-age populations, supporting data-driven planning and resource allocation.

EDUCATION

Ph.D., Engineering, University of California, Davis 2011

CERTIFICATIONS

- Large Language Models Professional Certificate, Databricks (2024)
- Creating Dashboards and Storytelling with Tableau, UC Davis (2024)
- Interpreting Machine Learning Models, Uplimit (2023)
- Deep Learning Specialization, DeepLearning.AI (2023)
- Data Scientist Certification, The Data Incubator (2022)